Mapping the Greatest Goal Scorer:
Alex Ovechkin's 924 Career Goals

Information Visualization Final Project

Data Source: NHL Public API (api-web.nhle.com) · Seasons: 2005–2025

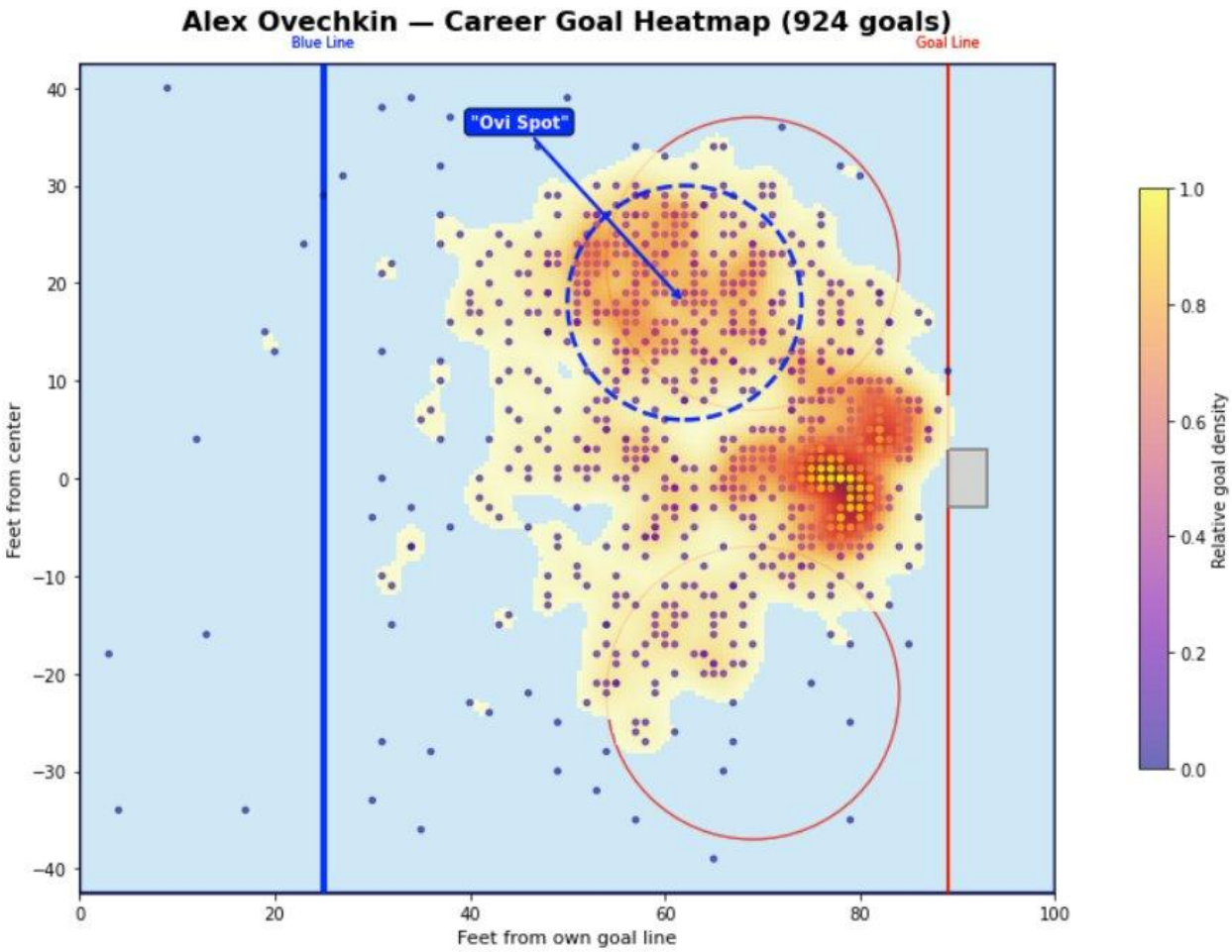

Figure 1. Career goal heatmap — all 924 regular-season goals (2005–2025)

Legend:

- **Yellow/red heatmap surface** — Continuous goal density via KDE; darker red = higher concentration of goals.
- **Purple to yellow dots** — Individual goals colored by local density (plasma colormap); yellow = highest density.
- **Dashed blue ellipse** — The "Ovi Spot," Ovechkin's signature left-circle scoring zone.
- **Blue vertical line** — The blue line (25 ft from center), marking the offensive zone boundary.
- **Red vertical line** — The goal line (89 ft from center), marking the front of the net.
- **Red circles** — Face-off circles in the offensive zone.
- **Gray rectangle** — The net.

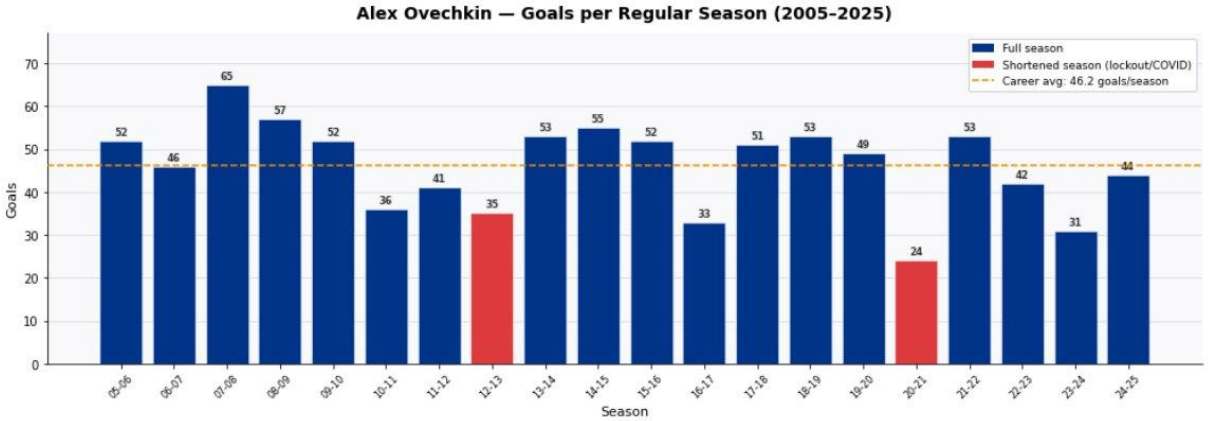

Figure 2. Goals per regular season (2005–2025)

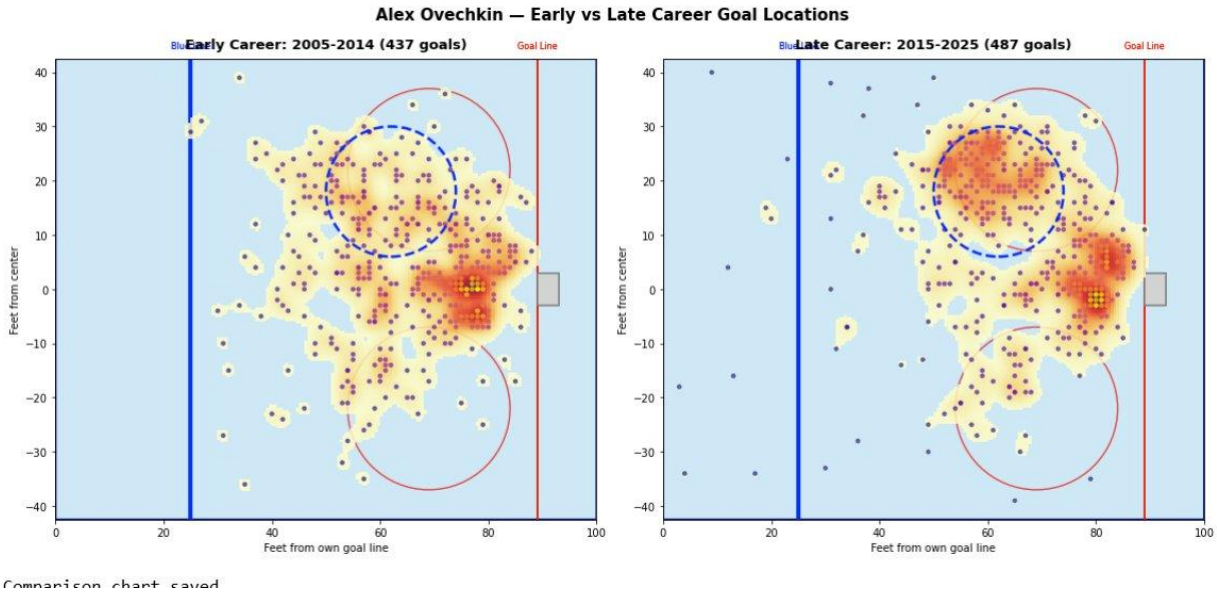

Figure 3. Early career (2005–2014) vs late career (2015–2025) goal locations

Key Findings:

- **The "Ovi Spot" is real and statistically proven -** The densest cluster in Figure 1 sits on the left circle of the offensive zone, roughly 15-25 ft from the goal. This location is where Ovechkin's one-timer off the left wing originates and it produces a disproportionate share of his career total. No other zone on the ice comes close in concentration besides the front of the net which is the most common place to score.

- **A secondary hotspot exists directly in front of the net -** A distinct high-density cluster appears near the front of the net. These goals are primarily deflections, rebounds, and tap-ins, showing that Ovechkin scores effectively in traffic as well as from distance. While Ovechkin clearly scored more goals from this part of the ice, this is also the most common place for hockey players to score.

- **His output has been remarkably consistent across two decades -** Figure 2 shows that in full 82-game seasons, Ovechkin scored between 41 and 65 goals, staying above his 46.2 career average in 13 of 18 full seasons. Even in his late 30s he remained close to or above average, a testament to unusual longevity at the elite level.
- **Shortened seasons are clear outliers -** The 2012–13 lockout season (35 goals in 48 games) and 2020–21 COVID season (24 goals in 45 games) are highlighted in red. On a per-game basis both seasons were consistent with his career rate, confirming the lower totals reflect fewer games rather than a drop in performance.
- **The Ovi Spot has become more concentrated in his later career -** Figure 3 directly compares his early career (2005–2014, 437 goals) against his late career (2015–2025, 487 goals). The left circle cluster is noticeably tighter and more intense in the late career panel, suggesting that Ovechkin has become slower and less athletic with age and has leaned even more heavily on his signature one-timer from that specific zone.
- **Overall shot distribution has narrowed with age -** The early career heatmap shows a broader distribution of goals across the offensive zone. The late career heatmap is more compact, with the two main clusters becoming more defined and dominant. This suggests a deliberate specialization over time, maximizing efficiency from his highest-percentage zones.

Data and Methods:

Data source - All goal location data was retrieved from the NHL's public REST API (api-web.nhle.com). For each of Ovechkin's 20 regular seasons (2005–06 through 2024–25), the player game log endpoint was queried to obtain individual game IDs. Each game's full play-by-play was then fetched from the gamecenter endpoint, and plays of type "goal" where the scoring player ID matched Ovechkin's (8471214) were extracted. The x and y coordinates embedded in each goal play represent the shot location in feet, measured from the center of the ice. A total of 924 goals with valid coordinates were collected.

Coordinate normalization - Because teams alternate which end they attack each period, raw coordinates can be positive or negative depending on direction of play. All goals were normalized to a single attacking direction by taking the absolute value of the x coordinate and mirroring the y coordinate accordingly, so the entire dataset represents Ovechkin shooting toward the right side of Figure 1.

Kernel density estimation - A continuous heatmap surface was generated using SciPy's gaussian_kde function with a bandwidth of 0.12. The KDE was evaluated on a 200×200 grid covering the offensive zone. Low-density regions below 5% of the maximum were masked so the ice background remains visible. The YlOrRd colormap was applied to the surface.

Individual goal coloring - Each goal was plotted as a scatter point and colored by its local KDE density value, normalized to the 0–1 range and rendered with the plasma colormap, allowing individual goals to be visible while reinforcing the density pattern.

Season bar chart - Goals were aggregated by season from the same dataset. Shortened seasons were identified by season code and highlighted in red. A career average line was computed across all seasons including shortened ones.

Tools - All data collection and visualization was implemented in Python using the requests, pandas, numpy, scipy, and matplotlib libraries.

Significance:

Alex Ovechkin surpassed Wayne Gretzky's all-time NHL goals record in 2024, cementing his place as the greatest goal scorer in hockey history. What makes his achievement analytically remarkable is not just the volume of goals but the spatial consistency with which they were scored. This project provides a comprehensive, data-driven portrait of that consistency plotting every single regular-season goal across two decades on a single rink diagram and pairing it with a longitudinal view of his season-by-season output. The "Ovi Spot" has become one of the most discussed concepts in modern hockey analytics. Teams have spent 20 years fully aware of exactly where Ovechkin scores from, yet the concentration of goals in that zone has barely diminished. Together, Figures 1 and 2 show this clearly. The spatial heatmap shows where, and the bar chart shows how consistently over time. The project demonstrates the power of combining spatial and temporal visualization for sports analytics and shows how even the most predictable patterns in elite sport can remain nearly impossible to stop.

Github Repository:

https://github.com/jwbailz32/ovi-goal-heatmap/tree/main